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CUET UG

Biology

Biotechnology and its Applications

Official Syllabus Unit IX · Class 12 · CUET UG 2027 Cycle Notes

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) Biology syllabus 2026, the current 2027 baseline, Unit IX)	Skills the questions test
Application in agriculture — tissue culture, GMOs, Bt cotton	Application of biotechnology in agriculture – tissue culture, genetically modified organisms, Bt cotton, RNA interference.	Matching cry genes to pests, tracing the protoxin route, naming GM goals and the standard crop examples.
Application in agriculture — RNA interference	Application of biotechnology in agriculture – tissue culture, genetically modified organisms, Bt cotton, RNA interference.	Explaining dsRNA silencing and the Meloidogyne incognita sequence of steps in the right order.
Biotechnology in medicine — insulin	Biotechnology in medicine – insulin, gene therapy and molecular diagnosis.	Chain sizes, proinsulin and the C-peptide, the E. coli route and the product name humulin.
Biotechnology in medicine — gene therapy	Biotechnology in medicine – insulin, gene therapy and molecular diagnosis.	The ADA deficiency story: which cells, which vector, why the treatment has to be repeated.
Biotechnology in medicine — molecular diagnosis	Biotechnology in medicine – insulin, gene therapy and molecular diagnosis.	Choosing what PCR, ELISA and a radioactive probe each detect, and how early.
Transgenic animals	Transgenic animals.	The definition, the five standard uses, and the Rosie milk figures.
Ethical issues	Ethical issues.	GEAC's role, why GM organisms are regulated, judging ethical statements neutrally.
Biopiracy and patents	biopiracy and patents.	Defining biopiracy, matching each patent case (Basmati, turmeric, neem) to its outcome.

EXAM PATTERN

CUET UG pattern (NTA official information bulletin and syllabus, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The Biology paper maps strictly to NCERT Class 12, and Unit IX closes with the applications block — the named examples here (Bt cotton, insulin, ADA, Rosie, Basmati) are the raw material most questions in this unit are built from.

Agriculture I — GM Crops and Bt Cotton

Picture a cotton field where the main pest — the bollworm, a caterpillar that tunnels into the bolls — keeps winning despite heavy spraying. In 1996 farmers planted a new kind of cotton that kills that caterpillar by itself, and pesticide use dropped sharply. That cotton, **Bt cotton**, is the single most-used example in this whole unit, so we take it slowly: what a GM crop is, where the killing ability came from, and why the same poison that kills the caterpillar never harms us.

What "genetically modified" really means

- **Genetically modified organism (GMO):** a plant, animal or microbe carrying a gene that humans deliberately added or altered — the gene is **foreign DNA** that now works inside the new host.
- **The gene does not care about species:** DNA works by the same code everywhere, so a gene from a bacterium can be read by a plant cell and still make its protein.
- **Making such an organism is "genetic modification":** the tools (plasmids, vectors, rDNA technology) are the machinery of the **principles** unit; here we look at what those tools **produced**.
- **Three terms, one family:** GMO (the organism), **transgenic** (carrying a foreign gene), and **rDNA technology** (the method) — questions mix them freely, so keep all three anchored.

Why plant breeders want GM crops

Goal	How genetic modification helps
More stress-tolerant plants	Crops that survive cold, drought, salt or heat where ordinary varieties die.
Less chemical pesticide	Pest-resistant crops kill the pest themselves — Bt cotton is the standard example.
Less post-harvest loss	Fruit and grain that spoil more slowly on the shelf and in storage.
Better food value	Higher nutrition — golden rice carries extra provitamin A; minerals can be raised too.
Medicine from plants	Plants turned into sources of usable proteins and vaccines — the "edible vaccine" idea.

- **Read the table as a list of benefits:** each row is one classic statement-based option pair — right benefit, wrong example, or right example, wrong benefit.

The Bt cotton story, step by step

- **The source bacterium:** *Bacillus thuringiensis* (that is what "Bt" stands for) is a common soil bacterium. During a certain stage it forms spores that carry **protein crystals**.
- **The crystals hold a poison for insects:** these **insecticidal proteins** are named **cry proteins**, and the genes that code for them are **cry genes** (the name comes from "crystal").
- **Each cry gene has its own target: cryIAc** protects cotton against the **cotton bollworms**; **cryIIAb** targets the **corn borer**. Gene-to-pest matching is the most-checked pairing in this section.
- **Scientists moved the gene into cotton:** the cry gene was inserted into the cotton plant's genome, so the plant now makes the bacterial protein in its own tissues.
- **The poison is stored in a switched-off form:** inside the plant the protein exists as **inactive protoxin crystals** — potent only after dissolving in an insect gut.

THE BT ROUTE IN FIVE LINES

1. Bt cotton carries the cry gene (cryIAc, cryIIAb).
2. The plant makes **inactive protoxin** crystals.
3. The insect eats the tissue; its **alkaline gut** dissolves the crystal.
4. The freed protoxin is **activated** and binds the midgut lining.
5. The gut wall develops pores, breaks down — the insect dies.

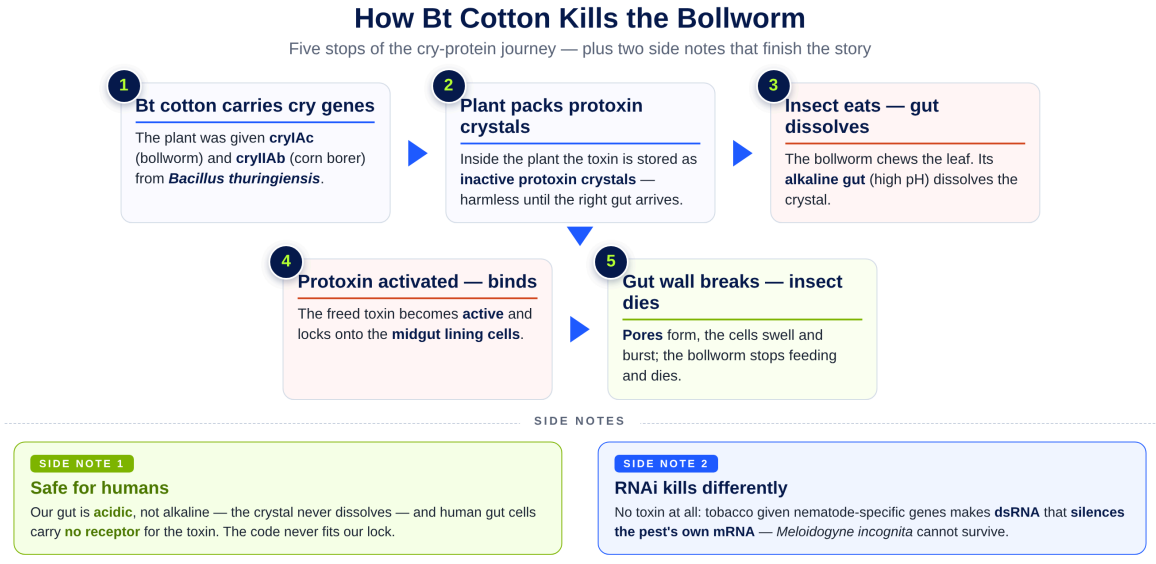


Figure 1: How Bt cotton kills the bollworm — the five stops, plus the two side notes (human safety and RNAi) that finish the story.

Why the same toxin never hurts humans

- **Our gut is acidic:** the crystal dissolves only in an **alkaline** (high-pH) gut. A human stomach is strongly acidic, so the protoxin simply never activates.

- **We have no receptor:** the activated toxin must first **bind** specific proteins on the insect's midgut cells. Human gut cells do not carry that binding site, so even activated toxin has nothing to attach to.
- **Both locks must fail:** the pest dies because its gut is both **alkaline** and **receptor-bearing**; we lack both. This "two-key" logic is the cleanest way to answer any "why is Bt safe" statement.
- **Specificity is the selling point:** a poison that kills one caterpillar family and nothing else is exactly what pest control wanted — spraying broad insecticides kills friend and foe alike.

REAL-WORLD ANALOGY — ONE KEY, ONE DOOR

Bt protoxin is like a key that is **locked inside a sealed envelope** (the crystal). Only the bollworm's alkaline gut is the right "opener" for the envelope, and only its midgut cells carry the matching keyhole. Humans are standing in front of a **different door with different weather** — acidic, and with no keyhole at all. The mapping is exact: envelope = protoxin crystal, opener = alkaline pH, keyhole = midgut receptor. Two conditions, both needed — which is why "Bt toxin harms humans" is always a wrong statement.

WORKED EXAMPLE — WHICH GENE FOR WHICH PEST?

Three fields, three complaints. Match each to its gene.

- **Field 1:** cotton bolls bored from inside by **bollworm** caterpillars → plant cotton with **cryIAc** — bollworms are its target.
- **Field 2:** maize leaves chewed by the **corn borer** → the matching gene is **cryIIAb**, not cryIAc.
- **Field 3:** tobacco roots knotted by the nematode *Meloidogyne incognita* → no cry gene helps; this is the **RNAi** case (next section).

Note the third case: Bt toxins kill **insects**; a nematode is a worm, so it needs the different tool. Keeping "insect → Bt" and "nematode → RNAi" apart settles a whole family of statement questions.

Agriculture II — RNA Interference and Tissue Culture

The first half of this section is about **silencing** a gene instead of adding one — no new protein is made, an existing message is simply stopped. The second half is about growing **thousands of plants from one cell**. Both are syllabus words in one line: RNA interference and tissue culture. They are also the easiest marks in this unit if you can retell the one NCERT experiment and the one set of culture steps in order.

RNA interference: silencing one gene

- **What mRNA is:** every gene works by first making an **mRNA** — a working copy of the recipe that the cell reads to build a protein. No mRNA read, no protein.

- **What RNAi is: RNA interference (RNAi)** is the silencing of a **specific** mRNA because a **complementary dsRNA** (double-stranded RNA) binds to it and stops it from being translated. The gene itself is untouched.
- **Where the dsRNA can come from:** naturally, from an infection by an RNA virus, or from **transposons** (mobile genetic elements) that copy themselves through an RNA intermediate. Biotechnologists simply **supply** the dsRNA on purpose.
- **Why it is a weapon against pests:** choose a gene the **parasite** needs, make dsRNA against it inside the host plant, and the parasite's own proteins stop being produced.

The NCERT experiment: nematode vs tobacco

- **The villain:** *Meloidogyne incognita*, the **root-knot nematode** — a tiny worm that infects tobacco roots and slashes yields.
- **The trick:** scientists introduced **nematode-specific genes** into the tobacco plant using **Agrobacterium vectors** (the standard DNA-delivery vehicle for plants).
- **The design detail that matters:** the DNA was introduced so the host produced **both sense and antisense RNA** — two RNA strands that are complementary to each other.
- **The two strands pair up:** sense + antisense RNA formed **dsRNA** inside the tobacco cells — and that dsRNA initiated RNAi.
- **The result:** the **specific mRNA of the nematode was silenced** by the dsRNA; the parasite could not survive in the transgenic host.

Step	What happens
1	Nematode-specific genes are inserted into tobacco via **Agrobacterium** vectors.
2	The host plant makes **sense RNA** and **antisense RNA** from the introduced DNA.
3	The two complementary strands pair to form **dsRNA** in the host cells.
4	dsRNA **initiates RNAi** and silences the matching nematode mRNA.
5	Essential nematode proteins are no longer made — **the parasite dies**.

- **Retell it in one breath:** genes in via Agrobacterium → sense + antisense RNA → dsRNA → mRNA silenced → nematode dead. Every statement-based question on RNAi is checking one of those five links.

Tissue culture: one cell, many plants

- **Totipotency:** the remarkable property of plants — **any** living cell can regenerate a whole plant, because it still carries the complete genome.
- **Explant:** the name for the piece you start with — **any part of a plant** (stem bit, leaf disc, meristem tip) taken out and grown in a test tube.

- **The nutrient medium:** the explant is grown on a sterile, special medium that supplies a **carbon source (sucrose, for example)**, **inorganic salts**, **vitamins**, **amino acids**, and **growth regulators** — auxins and cytokinins decide whether roots or shoots form.
- **Callus:** the soft, unorganised mass of cells that first grows out of the explant — the raw material from which organised shoots and roots later arise.
- **Micropropagation:** growing **thousands of plants** from one explant by tissue culture in a short time; every plantlet is a **somaclone** — identical to the original plant.
- **Virus-free bonus:** meristems (the growing tips) are generally free of viruses, so plants raised from **meristem culture** come out **virus-free** — the standard example for that statement.

Step	What happens in micropropagation
1	Take an **explant** from a healthy plant and sterilise its surface.
2	Place it on a sterile **nutrient medium** (sucrose, salts, vitamins, growth regulators).
3	Cells divide into an unorganised **callus** mass.
4	Change the regulator balance — **shoots**, then **roots**, form from the callus.
5	Move plantlets to soil; thousands of **somaclones** (or virus-free lines) result.

Somatic hybrids: fusing two plants

- **Protoplast:** a plant cell with its **cell wall removed** — a naked cell that can be fused with another.
- **Protoplast fusion:** two protoplasts (say, from two different varieties) are made to fuse; the combined cell regrows a wall and divides — a **somatic hybrid** plant develops.
- **The named example:** protoplasts of **tomato** and **potato** were fused to give the **pomato**. It is the classic illustration — and the classic trap, because the pomato **did not** combine all the desired traits for commercial use.
- **Why it matters:** somatic hybrids let breeders cross plants that sexual breeding cannot cross — the genes mix without seeds.

REAL-WORLD ANALOGY — PHOTOCOPIER AND CENSOR

Tissue culture is a photocopier for plants: feed one good page (the explant) and set the machine (the medium) — out come a thousand identical copies (somaclones). **RNAi** is a censor with one exact instruction: black out every copy of one sentence (the target mRNA) and let all others pass. The censor never enters the library (the DNA) — it works only on the circulating copies (mRNA). Keep the two jobs separate: copying cells is culture; censoring messages is RNAi.

WORKED EXAMPLE — SPOT THE WRONG STATEMENT

Four statements; find the false one.

- **(a)** RNAi uses complementary dsRNA to block a specific mRNA. — **True**; that is the definition.
- **(b)** The nematode experiment used Agrobacterium vectors to deliver the genes. — **True**; that is the standard plant delivery vehicle.
- **(c)** The introduced DNA made only antisense RNA in the host. — **False** — the plant made **both sense and antisense RNA**, and their pairing created the dsRNA that starts RNAi.
- **(d)** A callus is an unorganised mass of cells grown from an explant. — **True**; that is the standard definition.

The wrong statement hid a single word ("only") — statement questions in this unit turn on such small hinges: **both** strands, **complementary**, **specific**.

Medicine I — Human Insulin

Before 1983, every diabetic patient who needed insulin injections was, in a sense, borrowing it from a slaughtered animal. Then recombinant DNA technology changed the supply forever: bacteria were made to produce human insulin in fermentation tanks. This section follows that story, because it is built from a short list of checkable facts — two chains, two sizes, one cut-out peptide, one product name — and every one of them is a potential question.

Why the old route was a problem

- **The old source:** insulin was extracted from the **pancreas of slaughtered cattle and pigs** — a limited, grim supply line.
- **It worked, but not perfectly:** animal insulin is **almost** the same as human insulin — a few amino acids differ — and **some patients developed allergies** to the animal versions.
- **Demand only grew:** diabetes is common and lifelong, so a cheaper, safer, unlimited source was needed. The answer came from bacteria and plasmids.

What human insulin looks like

- **Two short chains:** mature human insulin consists of **chain A — 21 amino acids** — and **chain B — 30 amino acids**.
- **The chains are joined:** by **disulphide bonds** — sulphur-to-sulphur bridges between the chains. Both the count and the bond type are checkable facts.
- **It is not made directly:** insulin is first produced as **proinsulin**, a longer pro-hormone (like a prologue before a chapter). Proinsulin contains an extra stretch called the **C-peptide** that is **not present in mature insulin**.
- **Maturation cuts the C-peptide out:** enzymes remove the C-peptide, the chains fold, the disulphide bonds form — and mature insulin results.

INSULIN — THE NUMBERS TO CARRY

1. Chain A: 21 amino acids.

2. Chain B: 30 amino acids.
3. Joined by disulphide bonds.
4. Proinsulin = A + C-peptide + B.
5. C-peptide is cut out on maturation.

The rDNA route, step by step

- **Prepare the two recipes:** DNA sequences coding for the **A chain** and the **B chain** were prepared separately in the laboratory.
- **Load them into bacteria:** each sequence was inserted into the **plasmid DNA** of *E. coli* — the circular DNA vehicle that bacteria copy faithfully.
- **Chains produced separately:** the modified bacteria multiplied and produced the **A chains and B chains separately**.
- **Extract and join:** the two chains were **extracted**, then **combined** by creating **disulphide bonds** between them.
- **The product:** the result is **human insulin** made by microbes — marketed as **humulin**. This chain-by-chain route (Eli Lilly, 1983) was the first medicine of the rDNA era.

Point	Animal insulin (old route)	Humulin (rDNA route)
Source	Pancreas of slaughtered cattle/pigs	*E. coli* carrying the chain genes
Identity	Slightly different from human insulin	Same as human insulin
Side effects	Allergy in some patients	Largely avoided
Supply	Limited by slaughterhouses	Unlimited in fermentation tanks

- **A diagram in words:** gene for chain A → plasmid → *E. coli* → A chain; gene for chain B → plasmid → *E. coli* → B chain; A + B joined by disulphide bonds → **humulin**.

REAL-WORLD ANALOGY — SHIRT WITH A SPARE PANEL

Proinsulin is a finished shirt still carrying a **spare fabric panel** (the C-peptide) stitched between the sleeves. Before the shirt reaches the shop, the panel is unpicked — the sleeves (chains A and B) are already held together by their stitching (disulphide bonds). Two honest mappings to keep: the panel is **present only before maturation**, and the stitching, not the panel, joins the sleeves. So "insulin contains C-peptide" is wrong for **mature** insulin — the panel has been removed.

WORKED EXAMPLE — SORTING TRUE FROM FALSE ON INSULIN

Judge each statement.

- **(a)** Mature insulin has chain A of 21 amino acids and chain B of 30. — **True**; reversed sizes (30/21) are the standard wrong option.
- **(b)** Proinsulin is longer than mature insulin because of the C-peptide. — **True**; the extra stretch is cut out during maturation.
- **(c)** The rDNA route joined two random bacterial genes to make insulin. — **False**; specific **human** A-chain and B-chain sequences were used, produced separately in *E. coli* and then joined by disulphide bonds.
- **(d)** Humulin is insulin from human pancreases. — **False**; the name belongs to **human insulin produced by rDNA technology**.

Count the traps: sizes reversed, C-peptide's place, "random genes", and the source. That is the whole insulin section compressed into one card.

Medicine II — Gene Therapy and Molecular Diagnosis

Two different jobs share this section. **Gene therapy** tries to **fix** a broken gene by supplying a working copy. **Molecular diagnosis** tries to **find** a pathogen or mutation early, before symptoms speak. One is treatment, the other detection — and the exam's statement pairs love to swap their vocabulary, so we build both from their standard example.

Gene therapy: giving the working copy

- **The idea:** gene therapy is a set of methods where **genes are inserted into a person's cells and tissues** to treat a hereditary disease — a working copy is added to do the job the faulty gene cannot.
- **The standard example — ADA deficiency: adenosine deaminase (ADA)** is an enzyme needed for the immune system to work. Deficiency arises when the **gene for adenosine deaminase is deleted (faulty)**, and the result is a child with almost no immune defence (a severe combined immunodeficiency).
- **Existing half-measures:** in some children ADA deficiency can be helped by **bone marrow transplantation**, or by **enzyme replacement therapy** — injecting functional ADA. Both help, neither is the gene-level fix.

The 1990 therapy: the first of its kind

- **The patient:** a **4-year-old girl** with ADA deficiency — the world's first approved clinical gene therapy (1990).
- **Step 1 — collect: lymphocytes** (a type of white blood cell) were taken from the patient's blood and **grown in culture outside the body**.
- **Step 2 — deliver:** a **functional ADA cDNA** was introduced into those lymphocytes using a **retroviral vector** — a modified virus that carries the gene into the cells' chromosomes.
- **Step 3 — return:** the genetically corrected lymphocytes were **returned to the patient's body**, where they produced functional ADA.

- **The catch:** the therapy is **not a permanent cure** — lymphocytes do not live forever, so **periodic infusion** of corrected cells is needed. The permanent route would introduce the gene into cells that last a lifetime, such as **bone marrow cells at an early age**.

Question	Answer to remember
Which disease?	ADA deficiency — adenosine deaminase enzyme missing, immunity fails
Which cells?	Patient's own lymphocytes, grown outside the body
Which vector?	Retroviral vector carrying functional ADA cDNA
When?	1990 — the first clinical gene therapy
Why repeat?	Lymphocytes die over time; the fix lives only as long as the cells do

Molecular diagnosis: catch it early

- **The problem with old tests:** conventional diagnosis (serum and urine analysis) only **detects disease once symptoms appear** — often late.
- **PCR:** the **polymerase chain reaction** — **amplifies** a chosen piece of nucleic acid into millions of copies. Even a **very low concentration** of bacterial or viral DNA/RNA can be amplified and detected, long before symptoms.
- **ELISA:** a test built on **antigen–antibody interaction** — the pathogen's protein (antigen) is caught by a matching antibody, or the patient's antibodies are caught with a known antigen. It detects **proteins**, not genes.
- **Radioactive probe:** a **single-stranded DNA or RNA tagged with a radioactive molecule**; it **hybridises** (base-pairs) only to its complementary sequence. The sample is then exposed to an X-ray film — **autoradiography** — and the matched spot shows up.
- **The mutation check:** a mutated gene will **not appear** on the autoradiograph, because the probe finds no perfect complementarity — absence of the signal is itself the answer.

Tool	What it detects	One-line logic
PCR	Nucleic acid of the pathogen or mutant gene	Amplify the target a million-fold, then look
ELISA	Antigen or antibody (protein level)	Match the marker with an antibody
Probe + autoradiography	A specific DNA sequence, mutation-scan	Hybridise the tagged strand; read the film

Gene therapy vs molecular diagnosis, side by side

Point	Gene therapy	Molecular diagnosis
Purpose	Treat a hereditary disease	Detect a pathogen or mutation early
Acts on	The patient's cells (adds a working gene)	A sample from the patient (reads it)
Tools	Retroviral vector, corrected cells	PCR, ELISA, radioactive probes
Standard example	ADA deficiency, 1990	Early detection before symptoms

REAL-WORLD ANALOGY — REPAIR AND SMOKE ALARM

Gene therapy is the **repair crew**: it carries a new part (the working ADA gene) into the workshop (the lymphocytes) and puts the crew back to work. It does not hunt down the broken blueprint — the old gene simply stays. Molecular diagnosis is the **smoke alarm**: it looks for the earliest whiff of trouble (a few copies of pathogen DNA, one matching protein) long before the fire (symptoms) spreads. Repair changes you; the alarm only watches — and the watch comes first.

WORKED EXAMPLE — WHICH TOOL WOULD YOU PICK?

Three situations, one line each.

- **A newborn with repeated, severe infections** — suspect ADA deficiency: confirm at the gene level (probe/PCR) and treat by **gene therapy** — cells, vector, return.
- **A suspected viral infection with barely any virus in the blood** — use **PCR**: amplification turns a trace into a readable signal.
- **A blood sample that must be screened for a viral protein** — use **ELISA**: protein-level detection by antigen–antibody matching.

The pattern: **DNA → PCR/probe**, **protein → ELISA**, **fix the gene → therapy**.

Transgenic Animals

A transgenic animal is not just an animal with unusual traits — it is an animal whose DNA has been **deliberately edited to carry and express a foreign gene**. The gene came from somewhere else, and the animal's cells read it like their own. Mice dominate this field, one cow made headlines, and the syllabus lists exactly five uses — all five appear below, each with its named example.

What makes an animal transgenic

- **The definition:** an animal that has had its **DNA manipulated** to **possess and express an extra (foreign) gene** — the gene must be present **and** working.

- **The majority are mice:** about **95% of all transgenic animals are mice** — they breed fast, their genome is well mapped, and handling is standardised.
- **Why the numbers matter:** "95% mice" and "possess and express" are the two fact-checks this sub-topic hangs on — one number, one phrase.

The five uses of transgenic animals

Use	Standard example to attach to it
1. Study normal physiology and development	Transgenic rats carrying genes for **insulin-like growth factor** reveal how genes regulate growth and development.
2. Study of diseases	**Disease models**: transgenic animals made for **cancer**, **cystic fibrosis**, **rheumatoid arthritis**, **Alzheimer's** — human disease studied in a live body.
3. Producing useful biological products	Human protein **α-1-antitrypsin** (used to treat emphysema) produced in transgenic animals; **Rosie's** enriched milk (below).
4. Vaccine safety testing	**Polio vaccine** tested on **transgenic mice** — if approved by regulators, this could replace the use of monkeys.
5. Chemical safety testing	**Toxicity testing**: transgenic animals are **more sensitive to toxins**, so results arrive faster than with ordinary test animals.

- **Memory shape for the five: regulate** (genes), **sick** (disease models), **produce** (biologicals), **vaccinate** (vaccine safety), **poison** (toxicity). Five verbs, one row each — recite them in order and every list-type question is covered.

Rosie: the cow whose milk carried a human protein

- **The animal: Rosie**, the first **transgenic cow**, produced in **1997**.
- **The gene:** Rosie carried a **human gene** for **alpha-lactalbumin** — a protein of human milk.
- **The milk:** her milk contained the human protein at **2.4 grams per litre**.
- **Why it was news:** human-alpha-lactalbumin-enriched milk is a **nutritionally more balanced product for human babies** than ordinary natural cow milk.
- **Anchor the trio: Rosie — alpha-lactalbumin — 2.4 g per litre, 1997**. Swap any one of the three and the statement turns false — exactly how wrong options are written.

REAL-WORLD ANALOGY — A LIVING TEST PILOT AND FACTORY

A transgenic animal does two jobs at once. As a **test pilot**, it flies the human condition — a cystic-fibrosis mouse suffers the disease so treatments can be tried safely. As a **factory**, it produces goods on demand — Rosie's udder became a small dairy plant running on a human

recipe. The mapping is careful: the pilot works because the **same gene causes the same disease** in mouse and human; the factory works because a **mammal gland already makes milk proteins** — you only change the recipe.

WORKED EXAMPLE — MATCH THE USE TO THE EXAMPLE

Four examples; name the use.

- **Polio vaccine given to transgenic mice** → use **4**, vaccine safety testing (the monkeys-replacement line).
- **A mouse bred with the mutation behind cystic fibrosis** → use **2**, a **disease model** — the illness is studied in a live body.
- **Rats carrying insulin-like-growth-factor genes** → use **1**, normal physiology and development — how growth genes are regulated.
- **Rosie's milk at 2.4 g of human protein per litre** → use **3**, producing useful biological products (alpha-lactalbumin enrichment).

Every example is straight from NCERT's list — if you can match all four in under a minute, use-type questions are done.

Ethical Issues, Biopiracy and Patents

Biology now has the power to move genes between species — so who decides what may be released, and who pays for a seed, a leaf or a remedy that a community has known for centuries? The syllabus closes Unit IX with exactly these questions: a regulator (GEAC), a crime with a name (biopiracy), and three named patent cases. The facts are short, dated and easy to check — which is why they are worth perfect marks.

GEAC: India's gatekeeper for GM organisms

- **Who set it up:** the **Government of India** established the **Genetic Engineering Appraisal Committee (GEAC)**.
- **What it decides:** the **validity of GM research** and the **safety of introducing GM organisms for public services** — nothing GM reaches the public without this clearance.
- **Why regulation exists at all:** a released GM organism can breed, spread and out-compete wild relatives; the effects are hard to reverse. Oversight is therefore built into the law **before** release, not after.
- **Keep the label straight:** GEAC is an **Indian government committee** — not an international body. Confusing it with a world organisation is a standard wrong option.

Patents in plain words

- **What a patent is:** a **state-granted right** that lets an **inventor** stop others from making, using or selling the invention for a **fixed period** — in exchange, the invention is published for all to learn from.

- **The word doing the work is "inventor":** a patent rewards something **new**. A plant variety, seed or remedy in **public use for centuries** is **not** an invention — nobody can patent what was never theirs to invent.
- **Where abuse enters:** companies from wealthy nations (rich in money, poor in biodiversity) have patented products built on the **bio-resources and traditional knowledge** of developing nations — **without authorisation and without payment**. That abuse has a name.

Biopiracy: the definition and the three cases

- **Biopiracy:** the **use of bio-resources by multinational companies and other organisations without proper authorisation** from the countries and people concerned, and **without compensatory payment**.
- **The asymmetry behind it:** industrialised nations are **rich financially but poor in biodiversity**; developing nations are **rich in biodiversity and traditional knowledge** — so the flow of "free" raw knowledge has run one way.

Case	What happened	How it ended
Basmati rice	**RiceTec Inc.** (a US company) was granted a patent (1997) on basmati rice lines and grains developed by crossing Indian basmati with semi-dwarf varieties, and even sold the "new" rice as Basmati.	India filed objections; the **broad claims were withdrawn/cut down** (2001) — the original basmati stayed India's.
Turmeric	A US patent (1995) claimed turmeric's **wound-healing** property.	**CSIR** challenged it with **documented ancient texts**; the patent was **revoked in 1997**.
Neem	A European patent claimed neem oil's **fungicidal** property.	Challenged and **revoked (2000)** — traditional use again beat the patent.

- **The pattern across all three:** each patent tried to own a **traditional use** that was never new. Written records — ancient texts, documented practice — were the evidence that broke the "novelty" claim.

India's defences and the ethics habit

- **The Biological Diversity Act (2002):** India's law to **conserve biological diversity** and control how its bio-resources and knowledge are accessed and used — one line is all the syllabus needs here.
- **Documenting traditional knowledge:** written, dated, public records are what defeated the turmeric and neem patents — documentation is itself a defence.
- **Benefit sharing:** the fair answer to biopiracy is **authorisation plus compensation** — permission from, and payment to, the source community.

- **How such questions appear:** as **statement checks**, not opinions — "which body approves GM releases in India", "which patent was revoked and when", "what does biopiracy mean". Learn the labels and dates neutrally; no side-taking is ever required.

WORKED EXAMPLE — CORRECT THE ETHICAL STATEMENT

Three statements; fix each false one.

- **(a)** GEAC is an international committee under the WHO. — **False**: GEAC is set up by the **Government of India** to appraise GM research and releases.
- **(b)** The turmeric and neem patents are still valid. — **False**: turmeric was revoked in **1997**, neem in **2000**, after India's documented-evidence challenges.
- **(c)** A company may patent a community remedy if it improves it slightly. — **Not without permission and payment**: the underlying knowledge belongs to the community; unauthorised use is **biopiracy**, whatever the modification.

Notice how every fix is a **label, a body, a date** — this section rewards precision, not essays.

Key Facts & Terms Cheatsheet

KEY FACTS & TERMS CHEATSHEET — UNIT IX

Term / fact	Meaning in one line	Example / number
cryIAc / cryIIAb	Bt cry genes; each kills a specific insect pest	cryIAc: cotton bollworm; cryIIAb: corn borer
Protoxin activation	Inactive crystal turns lethal only in the insect's alkaline gut	Binds midgut cells → pores → insect dies
Human safety from Bt	Acidic human gut + no receptor — toxin never activates or binds	"Two-key" logic: pH and receptor both fail in us
RNAi	Complementary dsRNA binds a specific mRNA and blocks its translation	*Meloidogyne incognita* silenced in tobacco
RNAi delivery	Nematode-specific genes via **Agrobacterium** → sense + antisense RNA → dsRNA	Parasite cannot survive in the host
Totipotency	Any living plant cell can regenerate a whole plant	Explant → callus → plantlets
Micropropagation	Thousands of identical plants (somaclones) from one explant	Meristem culture gives **virus-free** plants
Protoplast fusion	Wall-less cells fused to make a somatic hybrid	Tomato + potato = pomato (traits not all kept)

Term / fact	Meaning in one line	Example / number
Insulin chains	A = 21 amino acids, B = 30, joined by disulphide bonds	51 residues in total, two chains
C-peptide	Extra stretch in **proinsulin**, cut out on maturation	Absent from mature insulin
Humulin	Human insulin by rDNA: chains made separately in *E. coli*, then joined	Eli Lilly, 1983 — first rDNA medicine
ADA deficiency	Adenosine deaminase gene faulty → immunity fails	1990: lymphocytes + ADA cDNA via retrovirus
PCR	Amplifies trace nucleic acid for detection	Finds pathogen DNA before symptoms
ELISA	Antigen–antibody test at protein level	Detects pathogen antigen or antibody
Radioactive probe	Tagged ssDNA/RNA hybridises to its complement; read by autoradiography	Mutated gene gives no film signal
Transgenic animals	Carry and express an extra foreign gene	About 95% are mice
Rosie	Transgenic cow with human alpha-lactalbumin in milk	2.4 g per litre, 1997
GEAC	Genetic Engineering Appraisal Committee — Indian body	Clears GM research and public releases
Biopiracy	Using bio-resources/traditional knowledge without authorisation or payment	Basmati (claims cut 2001); turmeric revoked 1997; neem revoked 2000

Common Errors and Special Cases

- **One habit before the table:** in this unit, most wrong statements fail on a **single hinge word** — active, mRNA, mature, alpha-lactalbumin, Indian, revoked. Read every statement with those hinges in mind.

COMMON ERRORS & SPECIAL CASES

The mistake students make	What is right	Why / the trap in one line
Saying Bt toxin is active as soon as it is made	It is an **inactive protoxin** until the insect's alkaline gut dissolves it	The activation step **is** the pest-specificity trick — and our safety
Saying RNAi cuts or edits the pest's DNA	RNAi **silences the mRNA**; the DNA stays untouched	dsRNA pairs with the working copy, never with the gene itself
Treating insulin as two genes cut apart at random	Insulin is made as **proinsulin with C-peptide**, cleaved on maturation; the rDNA route builds chains A and B separately, then joins them	Wrong chain sizes (30/21) and "random genes" are the two planted traps
Saying gene therapy removes the faulty gene	It **adds a working copy** (functional ADA cDNA) — the faulty gene remains	That is why corrected lymphocytes must be infused repeatedly
Attributing Rosie's milk protein to insulin	Rosie produced **alpha-lactalbumin**, 2.4 g per litre	Insulin belongs to the diabetes story; Rosie belongs to milk nutrition
Calling GEAC an international body	GEAC is set up by the **Government of India**	It appraises GM research and public releases **in India** only
Leaving turmeric/neem patents marked "active"	Both were **revoked** — turmeric 1997, neem 2000	Ancient-texts evidence killed the "novelty"; only Basmati's claims were narrowed, not revoked outright

- **Three checking habits to close with:**
  - **Pair every gene with its target:** cryIAc–bollworm, cryIIAb–corn borer, nematode-specific genes–*Meloidogyne*. No pair, no answer.
  - **Ask "which level":** DNA → probe/PCR; protein → ELISA; treatment → gene therapy. Level first, tool second.
  - **Date the ethics cases:** 1990 (ADA therapy), 1997 (Rosie, turmeric), 2000 (neem), 2001 (Basmati claims cut), 2002 (Biological Diversity Act).

The Whole Chapter on One Page

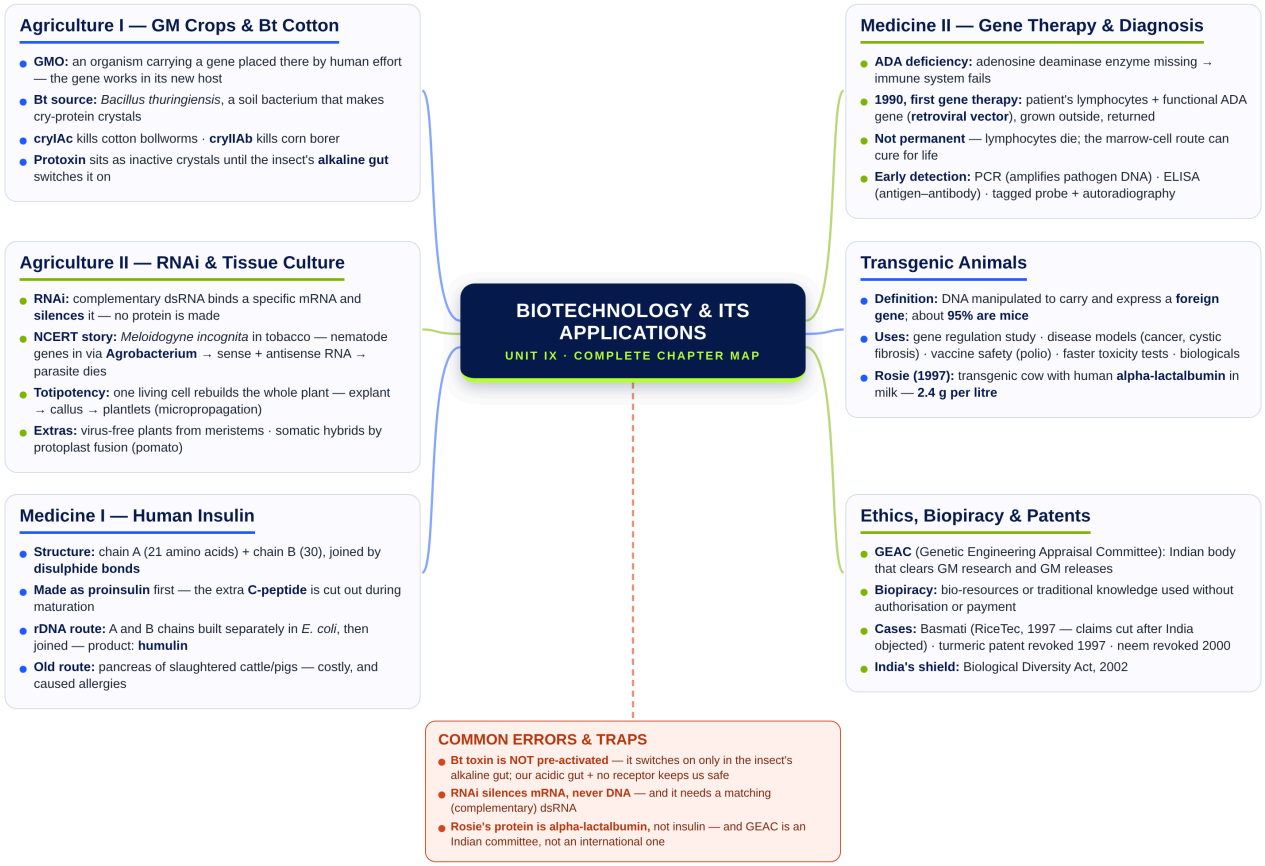


Figure 2: Whole chapter on one page — Unit IX as six branches, with the three traps that catch most students.

Practice Questions

Q1. A cotton plant carrying the **cryIAc** gene is protected chiefly against:

(A) corn borer
(B) root-knot nematode
(C) cotton bollworm
(D) rice weevil

Q2. Inside the cells of Bt cotton, the insecticidal protein is stored as:

(A) fully active toxin molecules
(B) inactive protoxin crystals
(C) double-stranded RNA
(D) antibody-like proteins

Q3. The Bt protoxin becomes lethal for the bollworm only when:

(A) sunlight ripens the cotton boll
(B) it meets the acidic human stomach
(C) soil bacteria chew the leaf
(D) the insect's alkaline gut dissolves the crystal

Q4. In the tobacco plants modified by RNA interference, the parasite whose mRNA was silenced was:

(A) *Meloidogyne incognita* **(B)** *Bacillus thuringiensis*
(C) *Agrobacterium tumefaciens* **(D)** *Plasmodium falciparum*

Q5. Mature human insulin consists of:

(A) chain A of 21 amino acids and chain B of 30, joined by disulphide bonds **(B)** chain A of 30 and chain B of 21, joined by peptide bonds
(C) one single chain of 51 amino acids **(D)** three chains with the C-peptide still inside

Q6. During the maturation of proinsulin into insulin, the stretch that is removed is:

(A) chain A **(B)** chain B
(C) the C-peptide **(D)** a disulphide bridge

Q7. In the 1990 gene therapy for ADA deficiency, the cells collected from the patient and grown in culture were:

(A) bone marrow stem cells **(B)** lymphocytes
(C) skin fibroblasts **(D)** red blood cells

Q8. ELISA detects a pathogen by means of:

(A) amplifying its nucleic acid **(B)** an antigen–antibody interaction
(C) autoradiography of a tagged probe **(D)** protoplast fusion

Q9. Rosie, the first transgenic cow (1997), produced milk containing:

(A) human insulin at 2.4 g per litre **(B)** the ADA enzyme at 2.4 g per litre
(C) human alpha-lactalbumin at 2.4 g per litre **(D)** cry protein at 2.4 g per litre

Q10. Which of these patents was revoked after India produced documented ancient-texts evidence of traditional use?

(A) the RiceTec Basmati patent **(B)** the neem patent
(C) the Bt cotton patent **(D)** the turmeric patent

Solutions

Sol. Q1. Gene-to-pest matching: **cryIAc** protects cotton against **cotton bollworms**; **cryIIAb** is the corn-borer gene; a nematode is fought by **RNAi**, not by cry genes.

Answer: (C)

Sol. Q2. In the plant the protein sits as **inactive protoxin crystals** — harmless until an insect gut dissolves them. Double-stranded RNA belongs to the RNAi story, and "fully active" reverses the activation logic that makes Bt specific.

Answer: (B)

Sol. Q3. Activation needs the **alkaline** gut of the insect: the crystal dissolves, the freed protoxin becomes active and binds the midgut lining. The human stomach is acidic, so option (B) describes exactly why Bt **never** harms humans.

Answer: (D)

Sol. Q4. The root-knot nematode is **Meloidogyne incognita** — nematode-specific genes were put into tobacco via Agrobacterium vectors, so *Agrobacterium* was the **delivery vehicle**, not the target; *Bacillus thuringiensis* is the Bt source bacterium.

Answer: (A)

Sol. Q5. Mature insulin: **chain A = 21 amino acids, chain B = 30**, joined by **disulphide bonds**. "One chain of 51" ignores the two-chain structure, and the C-peptide has already been removed from mature insulin.

Answer: (A)

Sol. Q6. Proinsulin is a longer pro-hormone: A chain + **C-peptide** + B chain. Maturation **cuts the C-peptide out**; chains A and B stay and hold the disulphide links.

Answer: (C)

Sol. Q7. The 1990 therapy used the girl's own **lymphocytes**: collected, grown in culture, given a functional ADA cDNA via a retroviral vector, then returned. Bone marrow cells are the **permanent** route, not the one used in 1990.

Answer: (B)

Sol. Q8. ELISA is built on **antigen–antibody interaction** — protein-level detection. Amplifying nucleic acid is PCR; a tagged probe read on film is autoradiography; protoplast fusion makes somatic hybrids.

Answer: (B)

Sol. Q9. Rosie carried a **human** gene for **alpha-lactalbumin** and her milk contained it at **2.4 g per litre** — a nutritionally better milk for human babies. Insulin and ADA belong to medicine; cry protein belongs to Bt plants.

Answer: (C)

Sol. Q10. The **turmeric** patent (US, 1995, wound-healing) was **revoked in 1997** after CSIR produced documented ancient texts. Neem was also revoked (2000) after a challenge;

the Basmati claims were **withdrawn/narrowed** (2001) rather than revoked through ancient-texts evidence.

Answer: (D)

BEFORE YOU CLOSE — THE 60-SECOND RECALL DRILL

Cover the right column, say the answer aloud, then check. If any row takes longer than three seconds, revisit its section.

Cue	Instant answer
cryIAc — which pest?	Cotton bollworms (cryIIAb takes the corn borer)
Where is Bt protoxin activated?	Only in the insect's alkaline gut — never in our acidic stomach
What does RNAi silence?	One specific mRNA, via complementary dsRNA — DNA untouched
The RNAi parasite and its vector	*Meloidogyne incognita*; genes delivered by *Agrobacterium*
Insulin chain sizes	A = 21 amino acids, B = 30, disulphide-joined
What is cut from proinsulin?	The C-peptide — absent from mature insulin
Humulin	rDNA human insulin — A and B chains made in *E. coli*, then joined (Eli Lilly, 1983)
The 1990 first	Gene therapy: ADA deficiency — lymphocytes + ADA cDNA via retrovirus
PCR / ELISA / probe — which level?	DNA / protein / DNA-sequence respectively
The Rosie trio	1997 · alpha-lactalbumin · 2.4 g per litre
Share of transgenic animals that are mice	About 95%
GEAC	Indian committee that appraises GM research and public releases
The patent endings	Turmeric revoked 1997 · neem revoked 2000 · Basmati claims cut 2001 · BDA 2002

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